

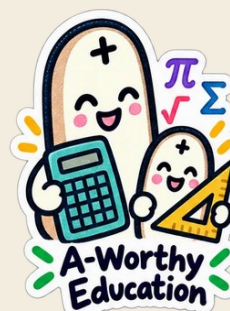


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O-Level Mathematics: Six Question Types, Six Frameworks

A named framework for each type of question on the O-Level Mathematics papers, with worked examples and the careless errors to check for.

LEVEL Secondary 1 to 4 · GCE O-Level Mathematics (Syllabus 4052)



How to use this guide

Each section names one question type, gives its framework as numbered steps, works an example and lists the errors the framework prevents. Every framework is the **Hit** step of the SHARP routine we use in class:

S See	H Hit	A Apply	R Refine	P Practise
Read the command word, the marks and the topic.	Pick the framework for that question type.	Write the working one operation per line.	Substitute back, check units, sense-check.	Rework the problem from memory days later.

The papers (SEAB 4052). Two papers, each 2 hours 15 minutes and 90 marks (50% each). Paper 1 has about 26 short questions; Paper 2 has 9 or 10 questions, and its last question applies mathematics to a real-world scenario. Relevant formulae are provided. SEAB states that omission of essential working will result in loss of marks. Give non-exact answers to 3 significant figures, or 1 decimal place for angles in degrees, unless the question says otherwise.

1 Word Problems: READ

Ratio, percentage, rate, speed and any problem written in words

Word problems test whether you can turn English into algebra. READ stops the commonest error: solving the equation correctly but answering the wrong question.

Steps

1. **Read** the question twice and underline what is being asked, not what is given.
2. **Extract** every given value and list the unknowns.
3. **Assign** variables: define x in words before you write an equation.
4. **Derive** the equation, solve it, then check the answer against the scenario: a speed cannot be negative, and a share of a whole cannot exceed 100%.

WORKED EXAMPLE

A man is four times as old as his son. In 6 years' time, he will be three times as old as his son. Find the son's present age.

R Asked for: the son's age now (not the man's).

E Now: man = $4 \times$ son. In 6 years: man = $3 \times$ son.

A Let the son's present age be x years. The man is $4x$ years old.

D $4x + 6 = 3(x + 6)$, so $4x + 6 = 3x + 18$, so $x = 12$. Check: 12 and 48 now; 18 and 54 in 6 years; $54 = 3 \times 18$. **Answer: 12 years.**

Errors this framework prevents

- Giving the value of the wrong unknown, such as the man's age.
- Writing an equation in letters that were never defined.
- An answer with no units, or one that makes no sense in the scenario.

2 Geometry: LABEL

Angles, triangles, polygons, circle properties, congruence and similarity

Geometry questions mark the reasoning as well as the calculation. LABEL makes you write a geometric reason beside every value you calculate.

Steps

1. **Label** the diagram: redraw it and mark every known length, angle, equal side, parallel line and right angle.
2. **Apply** the relevant properties: angle sum of a triangle, angles on a straight line, alternate angles, circle theorems, similarity ratios.
3. **Build** the solution chain one step at a time, with the reason beside each calculated value.
4. **Execute** each calculation cleanly.
5. **Link** back: check that the answer satisfies the original conditions.

WORKED EXAMPLE

Triangle PQR is isosceles with $PQ = PR$ and angle $QPR = 40^\circ$. QR is extended to S. Find angle PRS.

Working	Reason
Angle PQR = angle PRQ	Base angles of an isosceles triangle
Angle PRQ = $(180^\circ - 40^\circ) \div 2 = 70^\circ$	Angle sum of a triangle
Angle PRS = $180^\circ - 70^\circ = 110^\circ$	Angles on a straight line
Link: $40^\circ + 70^\circ = 110^\circ$, which agrees	Exterior angle = sum of the interior opposite angles

Errors this framework prevents

- A correct angle with no stated property.
- A chain that skips a step, so later values cannot be followed.
- Using a property whose condition you have not shown, such as alternate angles without parallel lines.

3 Graphs and Functions: PLOT

Linear, quadratic, cubic and exponential graphs

Graph questions test whether you understand the function behind the curve. PLOT makes you find the key features before you plot, because the examiner looks for them.

Steps

1. **Parameters**: identify the function type and note its key features. The sign of the leading coefficient tells you the shape.
2. **Locate** the key features before plotting: y-intercept, x-intercepts (roots), turning point, line of symmetry, asymptotes.
3. **Obtain** a table of values at strategic x-values, including the turning point and the intercepts.
4. **Trace** a smooth curve through the points. Label both axes, use a clear scale and mark every key feature.

WORKED EXAMPLE

Draw the graph of $y = x^2 - 2x - 3$ for values of x from -2 to 4 .

P Quadratic; the coefficient of x^2 is positive, so the curve is U-shaped with a minimum point.

L y -intercept $(0, -3)$. Roots: $(x - 3)(x + 1) = 0$, so $x = -1$ or $x = 3$. Line of symmetry $x = 1$, midway between the roots. Minimum point $(1, -4)$.

O x : $-2, -1, 0, 1, 2, 3, 4$ gives y : $5, 0, -3, -4, -3, 0, 5$.

T Draw a smooth U through the seven points; mark $(1, -4)$, the roots and the line $x = 1$.

Errors this framework prevents

- Joining the points with straight ruled lines instead of a smooth curve.
- Axes with no labels, or a scale that changes along an axis.
- Missing a key feature the question asks for, such as the turning point or a root.

4 Statistics: DATA

Mean, median, mode, interquartile range, standard deviation and probability

Statistics questions reward organised working, not mental arithmetic. Examiners give method marks for correct tabulation even when the final arithmetic slips, and DATA keeps that working on the page.

Steps

1. **Define** the measure the question requires: mean, median, mode, standard deviation, interquartile range or probability.
2. **Arrange** the data in a frequency table, an ordered list or a cumulative frequency table.
3. **Tabulate** the intermediate working: an $f \times x$ column and its total for the mean, cumulative frequencies for the median.
4. **Answer** in the context of the question, with units and the correct rounding.

WORKED EXAMPLE

The table shows how many books 20 students read last month. Find the mean and the median.

Books read (x)	0	1	2	3	4
Students (f)	3	7	6	3	1
$f \times x$	0	7	12	9	4
Cumulative frequency	3	10	16	19	20

Mean = total of $f \times x \div$ total of $f = 32 \div 20 = 1.6$ books.

Median: the 10th value is 1 and the 11th is 2, so the median is **1.5 books**.

Errors this framework prevents

- Dividing by the number of rows (5) instead of the total frequency (20).
- Taking the middle of the x column instead of the middle student.
- Rounding before the last step, or a probability outside 0 to 1.

5 Trigonometry: RULE

Right-angled and non-right-angled triangles, bearings, elevation and depression

Trigonometry goes wrong at the choice of method: SOH-CAH-TOA on a triangle with no right angle, or the sine rule where the cosine rule is needed. RULE settles the choice first.

Steps

1. **Right-angle check:** if the triangle has a right angle, use SOH-CAH-TOA.
2. **Use the correct formula:** with no right angle, use the cosine rule for two sides and the included angle, or the sine rule for a matching side-and-angle pair.
3. **Label** sides and angles by the standard convention: lowercase for sides, the matching uppercase for the opposite angles.
4. **Execute**, then sense-check: is the angle acute or obtuse as expected, is a bearing measured clockwise from north, and does the length fit the triangle?

WORKED EXAMPLE

In triangle ABC, $AB = 8$ cm, $AC = 6$ cm and angle $BAC = 50^\circ$. Find the length of BC. [3]

Wrong method: " $\cos 50^\circ = BC \div 8$, so $BC = 5.14$ cm." This assumes a right angle that is not there, so no method mark is available.

R No right angle. Two sides and the included angle are known, so use the cosine rule.

U $BC^2 = AB^2 + AC^2 - 2(AB)(AC) \cos A$

L $AB = 8$, $AC = 6$, $A = 50^\circ$

E $BC^2 = 64 + 36 - 96 \cos 50^\circ = 38.29\dots$, so **BC = 6.19 cm** (3 s.f.). Sense-check: BC lies between $8 - 6 = 2$ cm and $8 + 6 = 14$ cm, so it fits the triangle.

Errors this framework prevents

- SOH-CAH-TOA on a triangle with no right angle.
- The sine rule when only two sides and the included angle are known.
- A bearing measured anticlockwise, or rounding BC^2 before taking the square root.

6 Proofs and "Show That": SHOW

Any question that prints the result and asks you to prove it

The answer is printed in the question, so every mark goes to the reasoning. SHOW builds a stated reason into every line.

Steps

1. **State** the result you need to reach and write it at the top of your working.
2. **Hunt** for the theorems, identities or algebraic properties that lead there.
3. **Order** the chain from the given information to the target before you write.
4. **Write** each step formally, with the reason or theorem beside it. Never start from the result you are asked to show.

WORKED EXAMPLE

Show that $(n + 1)^2 - (n - 1)^2$ is a multiple of 4 for every integer n .

S Target: the expression equals $4 \times$ (an integer).

H Difference of two squares: $a^2 - b^2 = (a + b)(a - b)$.

O Factorise, simplify each bracket, then conclude.

W $(n + 1)^2 - (n - 1)^2 = [(n + 1) + (n - 1)][(n + 1) - (n - 1)]$ (difference of two squares) $= (2n)(2) = 4n$. Since n is an integer, $4n$ is a multiple of 4.

Errors this framework prevents

- Starting from the printed result and working backwards.
- Steps with no stated reason.
- Stopping at 4n without the concluding sentence.

7 The Seven Careless Errors

Check every script for these before you hand it in

Careless error	The fix
1. Working before decoding the question: the wrong form, or the wrong accuracy.	Underline the command word (solve, simplify, factorise, express, show); circle the required form and accuracy. Use READ for word problems.
2. Doing steps in your head and losing the method marks.	Write one operation per line, nothing skipped.
3. SOH-CAH-TOA on a non-right-angled triangle.	RULE: right-angle check first.
4. Rounding too early, and dropping the units.	Keep full calculator precision; round once, at the end; write the unit every time.
5. Geometry answers with no reasons.	LABEL: a stated property on every line.
6. Statistics by mental arithmetic.	DATA: build the table and show the totals.
7. Freezing on "show that" and proof questions.	SHOW: state the target, then build the chain.

Summary: every framework on one page

Question type	Framework	Steps
Word problems	READ	Read twice and underline what is asked; Extract values and unknowns; Assign variables in words; Derive, solve and check against the scenario.
Geometry	LABEL	Label the diagram; Apply the properties; Build the chain with a reason per value; Execute; Link back to the conditions.
Graphs and functions	PLOT	Parameters (type and shape); Locate intercepts, turning point, symmetry, asymptotes; Obtain a table of values; Trace a smooth, labelled curve.
Statistics	DATA	Define the measure; Arrange the data in a table; Tabulate the working ($f \times x$, cumulative frequency); Answer in context with units.
Trigonometry	RULE	Right-angle check; Use the correct formula (SOH-CAH-TOA, cosine rule or sine rule); Label by convention; Execute and sense-check.
Proofs and "show that"	SHOW	State the target; Hunt for theorems and identities; Order the chain; Write each step with its reason.
Every question	Final check	Command word and form; one operation per line; full precision until the end; 3 s.f. (angles 1 d.p.) unless told otherwise; units.

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